

MSDS 534 Fall 2025: Project Guideline

October 25, 2025

Instructions

You will, in a group of 1-4 students, select one of the papers listed and read it carefully. You will evaluate the methods described in the paper on either real or simulated data, by either implementing the methods yourself or using an existing software package; you are recommended to implement the methods yourself in certain papers. You may find good sources of data in the following sources:

- UCI repository: <https://archive.ics.uci.edu/ml/index.php>
- Kaggle: <https://www.kaggle.com/datasets>

You may propose a paper separate from the list if you get prior approval from the instructor; the paper must be related to one of the course topics. Or, you may propose a data analysis project of your own in which you apply machine learning methods to a specific data set. You must get prior approval for this as well.

Deliverables

The deliverable is a 4-6 page report. (If the report is too long, points will be deducted. Bibliography and figures are not included in page limit.)

The report should contain:

1. A description of the problem, including a precise mathematical formulation;
2. A description of the models, methods and algorithms, with well-defined notation;
3. Empirical evaluation of the methods using appropriate, well-defined metrics;
4. Interpretation of the empirical results;
5. Discussion of the advantages and disadvantages of the model/method/algorithm.

The report should cite sources appropriately. If you have questions about how to reference papers, please ask instructor. The report should also be well-structured and read well as a cohesive whole. We will also have a final presentation which is about 10min either in class or pre-recorded then upload on Canvas.

Key Dates

* Each of you have to submit the proposal, abstract and write-up separately on Canvas.

1. November 7: select paper or propose project. **No more than 2 groups per paper.**
2. November 21: submit abstract for project
3. December 17: project write-up due.

Suggested Project Ideas:

Papers

- Embeddings
 - Word2Vec ([link](#)) (Mikolov et al., 2013)
 - Exponential family embeddings ([link](#)) (Rudolph et al., 2016)
- SGD:
 - Hardt, M., Recht, B., & Singer, Y. (2016, June). Train faster, generalize better: Stability of stochastic gradient descent. In International conference on machine learning (pp. 1225-1234). PMLR.
 - Fine-grained analysis of stability and generalization for stochastic gradient descent, Yunwen Lei et al, 2020 ICML
- Deep models:
 - Deep exponential families ([link](#)) (Ranganath et al., 2015)
 - Toward better generalization bounds with locally elastic stability, Zhun Deng et al, 2021 ICML
- Large language models:
 - Fine-tuning with LORA ([link](#)) (Hu et al., 2021) (e.g., you can fine-tune a Gemma-2b model with LORA)
 - Were RNNs all we needed? ([link](#)) (Feng et al., 2024) (implement a small version of one of their RNNs)
 - CLIP (language + vision) ([link](#)) (Radford et al., 2021)
- Latent variable modeling / generative modeling:
 - MFCVAE ([link](#)) (Falck et al., 2021)
 - VQVAE ([link](#)) (Van Den Oord et al., 2017)
 - Diffusion models ([link](#)) (Ho et al., 2020)
- Conformal inference
 - Beyond iid/exchangeability ([link](#)) (Barber et al., 2023)

- Times series data (link) (Oliveira et al., 2024)
 - Conformal inference and fairness (link) (Romano et al., 2020)
 - Integrating uncertainty awareness into conformalized quantile regression (link) (Rossellini et al., 2024)
 - Aolaritei, L., Wang, Z. O., Zhu, J., Jordan, M. I., & Marzouk, Y. (2025). Conformal Prediction under Levy-Prokhorov Distribution Shifts: Robustness to Local and Global Perturbations. arXiv preprint arXiv:2502.14105.
 - Qiu, H., Dobriban, E., & Tchetgen Tchetgen, E. (2023). Prediction sets adaptive to unknown covariate shift. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 85(5), 1680-1705.
 - Ai, J., & Ren, Z. (2024). Not all distributional shifts are equal: Fine-grained robust conformal inference. arXiv preprint arXiv:2402.13042.
 - Zargarbashi, S. H., Akhondzadeh, M. S., & Bojchevski, A. (2024). Robust yet efficient conformal prediction sets. arXiv preprint arXiv:2407.09165.
 - Clarkson, J., Xu, W., Cucuringu, M., & Reinert, G. (2024). Split conformal prediction under data contamination. arXiv preprint arXiv:2407.07700.
 - Plassier, V., Kotelevskii, N., Rubashevskii, A., Noskov, F., Velikanov, M., Fishkov, A., & Panov, M. (2024, April). Efficient conformal prediction under data heterogeneity. In *International Conference on Artificial Intelligence and Statistics* (pp. 4879-4887). PMLR.
 - Yang, Y., & Kuchibhotla, A. K. (2025). Selection and aggregation of conformal prediction sets. *Journal of the American Statistical Association*, 120(549), 435-447.
 - Hegazy, M., Aolaritei, L., Jordan, M. I., & Dieuleveut, A. (2025). Valid Selection among Conformal Sets. arXiv preprint arXiv:2506.20173.
 - Bai, T., & Jin, Y. (2024). Optimized conformal selection: Powerful selective inference after conformity score optimization. arXiv preprint arXiv:2411.17983.
 - Liang, R., Zhu, W., & Barber, R. F. (2024). Conformal prediction after efficiency-oriented model selection. arXiv preprint arXiv:2408.07066.
 - Liang, Z., Zhou, Y., & Sesia, M. (2023, July). Conformal inference is (almost) free for neural networks trained with early stopping. In *International Conference on Machine Learning* (pp. 20810-20851). PMLR.
- Testing
 - Permutation-free independence test (link) (Shekhar et al., 2023)
 - Permutation tests using arbitrary permutation distributions (link) (Ramdas et al., 2023) (run experiments comparing these tests to the permutation tests in class)
 - Faster conditional randomization tests (link) (Liu et al., 2022)
 - Asymptotic distribution-free independence test for high-dimensional data (link) (Cai et al., 2024)
 - Model-free prediction test with application to genomics data (link) (Cai et al., 2022)

- Multiple testing
 - Online false discovery rate control (link) (Ramdas et al., 2018)
 - A unified treatment of multiple testing with prior knowledge using the p-filter (link) (RAMDAS et al., 2019)
 - A general interactive framework for FDR control under structural constraints (link) (Lei et al., 2020)
 - Knockoffs (Barber and Candès, 2015)
 - Tian, Y., Zhao, Z., & Sun, W. (2025). Conformalized Multiple Testing under Unknown Null Distribution with Symmetric Errors. arXiv preprint arXiv:2509.04231.
 - Zhang, T., Lee, H., & Lei, J. (2024). Winners with confidence: Discrete argmin inference with an application to model selection. arXiv preprint arXiv:2408.02060.
- Interpretability
 - Lassonet (Lemhadri et al., 2021)
 - Have We Learned to Explain?: How Interpretability Methods Can Learn to Encode Predictions in their Interpretations (link) (Jethani et al., 2021)
 - Variable Importance Inference in Classification (link) (Wang et al., 2024)
 - Impossibility theorems for feature attribution (link)(Bilodeau et al., 2024) (summarize key ideas, conduct experiments similar to paper)

References

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